(d) Two plants are selected at random from Plantation A.

The students are asked to calculate the probability of selecting a plant that is less than 150 cm and another plant that is at least 150 cm.

(i) Alice says that the above probability is calculated using $\frac{8}{21}$ and $\frac{13}{21}$.

Explain why she is wrong.

Answer

[1]

(ii) Calculate the above-mentioned probability.

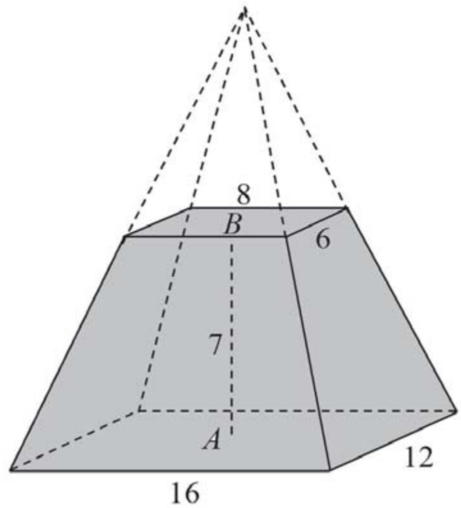
Answer ______________________ [2]

(e) 30 plants are planted in Plantation B on the same day. The range of height is 24 cm, the interquartile range is 10 cm and the median is 152 cm. Compare the height in Plantation A and B in two ways.

Answer

[2]

9 The frustum shown in the diagram is a metal solid obtained by cutting off the top part of a pyramid horizontally.
The dimension of the top rectangular plane is 8 cm by 6 cm.
The dimension of the bottom rectangular plane is 16 cm by 12 cm.
The height, AB, of the frustum is 7 cm where A is the center of the base.



(a) Explain why the height of the pyramid that is cut off is 7 cm.

Answer

[2]

(b) Show that the volume of the frustum is 784 cm^3.

Answer

[2]

(c) Assuming negligible loss in the process of melting, how many spherical balls (of radius 2 cm) can be formed if the frustrum is melted?

Answer ________________________ [2]

(d) 3 of the above frustums are to be packed into a rectangular box. To save cost, the seller should keep the amount of material of the rectangular box to the minimum.

Albert claims that the dimension should be 30 cm by 16 cm by 7cm (Box *A*).
Betty claims that the dimension should be 38 cm by 12 cm by 7cm (Box *B*).
Carol claims that the dimension should be 21cm by 16 cm by 12cm (Box *C*).

(i) Explain which box will be too small.

Answer

[1]

(ii) Which of the other two boxes would use the minimum amount of material?

Answer ________________________ [3]